

1. A relation from A to B is:

- **Definition:** A relation from a set A to a set B is any subset of the Cartesian product $A \times B$, i.e., all ordered pairs (a, b) where $a \in A$ and $b \in B$.
 - **Answer:** b) Subset of $A \times B$
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2. A function is a relation in which:

- **Definition:** A function assigns **exactly one output in B** for each input in A.
 - **Answer:** b) Every element of A maps to exactly one element of B
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3. Domain of a function contains:

- **Definition:** The domain is the set of **all input values** for which the function is defined.
 - **Answer:** d) Input values for which function is defined
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4. Range of a function is always a subset of:

- **Definition:** The range is the set of actual outputs. The codomain is the set of possible outputs. Therefore, $\text{range} \subseteq \text{codomain}$.
 - **Answer:** b) Codomain
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5. Reflexive relation on $A=\{1,2\}$ is:

- **Definition:** Reflexive means every element relates to itself. For $A = \{1,2\}$, reflexive pairs are $(1,1)$ and $(2,2)$.
 - **Answer:** b) $\{(1,1), (2,2)\}$
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6. A symmetric relation satisfies:

- **Definition:** Symmetric means if (a,b) is in the relation, then (b,a) must also be in the relation.
 - **Answer:** b) $(a,b) \Rightarrow (b,a)$
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7. A transitive relation satisfies:

- **Definition:** Transitive means if (a,b) and (b,c) are in the relation, then (a,c) must also be included.
 - **Answer:** b) $(a,b), (b,c) \Rightarrow (a,c)$
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8. An equivalence relation is:

- **Definition:** Equivalence relation satisfies **Reflexive + Symmetric + Transitive**.
 - **Answer:** d) Reflexive + Symmetric + Transitive
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9. Which is a function?

- **Check:** A function cannot map the same input to multiple outputs.
 - a) $(1 \rightarrow 2, 1 \rightarrow 3)$ ❌ Not a function
 - b) $(1 \rightarrow 1, 2 \rightarrow 2)$ ✅ Valid function
 - c) $(2 \rightarrow 2, 2 \rightarrow 3)$ ❌ Not a function
 - d) $(1 \rightarrow 1, 1 \rightarrow 2)$ ❌ Not a function
 - **Answer:** b) $\{(1,1),(2,2)\}$
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10. One-one function means:

- **Definition:** One-one (injective) means each input has a unique output and no output repeats.
 - **Answer:** b) Input unique $\rightarrow$ output unique
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11. Onto function means:

- **Definition:** Onto (surjective) means **range = codomain**. Every element of codomain must be an output.
 - **Answer:** b) Range = Codomain
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12. Constant function is always:

- **Definition:** A constant function gives the **same output for all inputs**, so it is **many-one** (multiple inputs $\rightarrow$ same output).
 - **Answer:** c) Many-one
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13. $f(x)=|x|$ is:

- **Analysis:** Absolute value function gives the **same output for $\pm x$** , so multiple inputs can map to the same output.
 - **Answer:** c) Many-one
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14. Number of functions from $A=\{1,2\}$ to $B=\{3,4,5\}$:

- **Formula:** Number of functions = $|B|^{|A|} = 3^2 = 9$.

- Answer: c) 9
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15. $f(x)=1/x$ is not defined at:

- Reason: Division by zero is undefined.
 - Answer: a) $x=0$
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16. Domain of $\sqrt{x}$:

- Reason: Square root requires $x \geq 0$.
 - Answer: b) $x \geq 0$
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17. Range of $f(x)=x^2$ for $x \in \mathbb{R}$:

- Reason: Squaring any real number ≥ 0 , so range is $x \geq 0$.
 - Answer: c) $x \geq 0$
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18. $R=\{(1,2),(2,1)\}$ on $A=\{1,2\}$ is:

- Check: Symmetric if $(a,b) \Rightarrow (b,a)$. Here $(1,2)$ and $(2,1)$ exist $\rightarrow$ symmetric. Not reflexive (missing $(1,1),(2,2)$), not transitive.
 - Answer: b) Symmetric
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19. Composite function $(g \circ f)(x)$ equals:

- Definition: $(g \circ f)(x) = g(f(x))$
 - Answer: d) $g(f(x))$
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20. A function is invertible iff it is:

- Reason: A function is invertible iff it is **bijective** (both one-one and onto).
 - Answer: d) Bijective
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21. Universal relation on set A is:

- Definition: Universal relation contains **all possible pairs** from $A \times A$.
 - Answer: a) $A \times A$
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22. Identity relation contains:

- Definition: Identity relation contains **all pairs of the form (a,a)**.
 - Answer: a) All (a,a)
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23. Which is many-one?

- Check: Many-one means multiple inputs can map to the same output.
 - $y = x^2$ ✓ Many-one
 - $y = x$ ✗ One-one
 - $y = \sin x$ ✗ Many-one (periodic but still many-one is possible)
 - Answer: b) $y = x^2$
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24. $f(x) = \sin x$ is:

- Reason: Sine is **not one-one** over $\mathbb{R}$ (repeats values periodically).
 - Answer: c) Not one-one
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25. Domain of $\log x$:

- Rule: $\log(x)$ is defined only for $x > 0$.
 - Answer: a) $x > 0$
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26. Range of $\log x$:

- Reason: $\log(x)$ can take any real value.
 - Answer: c) $\mathbb{R}$
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27. $R = \{(1,1), (2,2), (1,2), (2,1)\}$ is:

- Check: Reflexive: (1,1), (2,2) ✓
 - Symmetric: (1,2) and (2,1) ✓
 - Answer: c) Both
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28. Number of relations on $A = \{1,2,3\}$:

- Formula: Number of relations = $2^{(n^2)}$, $n = |A| = 3 \rightarrow 2^9 = 512$.
 - Answer: c) 512
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29. Which is equivalence?

- **Check:** Equivalence = Reflexive + Symmetric + Transitive.
 - $a = b$  Equivalence
 - $a < b$  Not symmetric
 - a divides b  Not always symmetric
 - **Answer:** a) $a = b$
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30. $f(x)=x^3$ is:

- **Check:** x^3 is one-one (strictly increasing) and onto $\mathbb{R} \rightarrow$ bijective.
 - **Answer:** d) All (one-one, onto, bijective)
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31. Domain of $1/\sqrt{x}$:

- **Reason:** $\sqrt{x}$ requires $x \geq 0$, $1/\sqrt{x}$ requires $x \neq 0 \rightarrow x > 0$
 - **Answer:** a) $x > 0$
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32. Domain of $\sqrt{4-x^2}$:

- **Reason:** Under root $\geq 0 \rightarrow 4-x^2 \geq 0 \rightarrow -2 \leq x \leq 2$
 - **Answer:** c) $-2 \leq x \leq 2$
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33. $f(x)=|x|$ is not invertible because:

- **Reason:** Not one-one ($|-a| = |a|$) $\rightarrow$ many-one.
 - **Answer:** c) It is many-one
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34. $[x]$ represents:

- **Definition:** Floor function = greatest integer $\leq x$
 - **Answer:** b) Greatest integer $\leq x$
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35. "Parallel to" is:

- **Reason:** Parallel lines are **reflexive** (line $\parallel$ itself), **symmetric** (if $A \parallel B$ then $B \parallel A$), and **transitive** (if $A \parallel B$ and $B \parallel C$ then $A \parallel C$) $\rightarrow$ equivalence relation.
 - **Answer:** c) Equivalence
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36. Range of $\sin x$:

- Fact: $\sin x \in [-1, 1]$
 - Answer: b) $[-1, 1]$
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37. $f(x) = \cos x$ is:

- Reason: Cos is periodic $\rightarrow$ not one-one, many-one.
 - Answer: b) Many-one
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38. $f(x) = x^2$ is one-one on:

- Reason: Squaring is one-one **only on** $x \geq 0$ or $x \leq 0$, not on $\mathbb{R}$.
 - Answer: a) $x \geq 0$
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39. $\tan x$ is undefined at:

- Reason: $\tan x = \sin x / \cos x \rightarrow \cos x = 0 \rightarrow x = \pi/2 + n\pi$.
 - Answer: b) $\pi/2$
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40. $\log(0)$ is:

- Reason: $\log(0)$ is undefined because $10^? = 0$ has no solution.
- Answer: b) Undefined